

Abhinav Karnatak

Full-Stack Developer | Applied AI & Agentic Engineering

abhinavkarnatak2004@gmail.com • +91-7310988194 • [linkedin.com/in/abhinavkarnatak](https://www.linkedin.com/in/abhinavkarnatak)
github.com/abhinavkarnatak-dev • abhinavkarnatak.com

Full-stack developer with production experience across React/Next.js, Node.js, and microservices architecture. Currently extending that foundation into applied AI and agentic systems - building with LangChain, LangGraph, and RAG - while continuing to ship full-stack, production-style projects rather than isolated notebooks.

SKILLS

AI/LLM Engineering: LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Agentic Workflows, LLM Evaluation, Guardrails, Prompt Engineering, LLM API Integration

Programming Languages: JavaScript, TypeScript, C++, Python

Frontend Technologies: React.js, Next.js, React Native, Tailwind CSS, Framer Motion, Progressive Web Apps (PWA), Figma

Backend & Databases: Node.js, Express.js, MongoDB, PostgreSQL, Prisma, Redis, RabbitMQ

DevOps and Cloud: Docker, AWS (EC2, S3), Nginx

Libraries & Integrations: Socket.io, Gemini API, Razorpay, Scrapy, Recharts

PROJECTS

YapIt: Real-Time Microservices Chat Application

- Built using **Next.js, Node.js, Socket.io, and Redis**, YapIt is a highly scalable full-stack web app where users can engage in seamless real-time conversations.
- Engineered a decoupled **Microservices Architecture** (User, Chat, and Mail services) utilizing **RabbitMQ** for reliable inter-service messaging and asynchronous task handling like global account data cleanup.
- Containerized the entire ecosystem with **Docker Compose** and deployed it to an **AWS EC2** instance, utilizing **Nginx** reverse proxy to securely manage API routing, continuous WebSocket traffic, and AWS S3 file-sharing.

Revuea: Empowering Honest Feedback with Anonymity

- Developed a full-stack web application using **React.js, Node.js, and PostgreSQL** (via **Prisma ORM**) to securely collect, store, and manage anonymous user feedback.
- Implemented a secure authentication flow using **OTP-based signups** and built a dashboard utilizing **Recharts** to visualize feedback data efficiently.
- Integrated the **Gemini AI API** to instantly process large volumes of user feedback into readable text summaries, while also adding **CSV export** for external data analysis.

PixScribe: SaaS Text-to-Image Generator Web App

- Built using **React.js, Node.js, Express, and MongoDB**, PixScribe is a full-stack web app that lets users generate AI-based images from text prompts.
- Engineered a credit system linked to a **Razorpay** integration, handling secure payment processing and automatically updating user account balances upon successful transactions.
- Designed a fast, highly responsive user interface with **Tailwind CSS** and integrated **Framer Motion** to deliver smooth layout animations and state transitions.

AdaptSense: Empowering the Visually & Hearing Impaired with AI

- Developed a cross-platform mobile application using **React Native (Expo)** to improve daily accessibility for visually and hearing-impaired users.
- Constructed a flexible media pipeline (audio, image, text, braille inputs) routing to visual/auditory outputs, using **Gemini APIs** to process and translate 13 language-media combinations with high accuracy.

WORK EXPERIENCE

Analyst (Software Engineer)

Sep 2025 - May 2026

Capgemini

Mumbai, Maharashtra

- Assisted in troubleshooting network and connectivity issues, gaining hands-on exposure to enterprise server environments (Linux/Windows) while ensuring timely resolution of assigned tickets.
- Collaborated with the team to monitor system health and ensure smooth daily operations.

Data Engineer Intern

Jun 2024 - Jul 2024

HERE Technologies

Remote

- Built custom Python web crawlers using **Scrapy** to scrape store locations from retail websites with varying HTML layouts, scheduling automated runs that cut manual effort by 30%.
- Documented the workflow and codebase to make it easier to maintain and update in the future.

EDUCATION

B.Tech Computer Science and Engineering

Aug 2021 - May 2025

University Of Petroleum and Energy Studies

- Served as the **Student Placement Representative (SPR)**, coordinating communication between the placement cell and students during campus recruitment activities.